ERDŐS PROBLEM 634: OBSTRUCTIONS, SCOPE LIMITS, AND CLOSED ROUTES

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Abstract. This note collects the negative results of the base-β programme: the exact scope of each theorem in the companion, the mechanisms proved unable to close the remaining case, and the routes closed with their precise failure points. Nothing here is needed to read [1] or [2]. It exists so that closed routes are not retried, and so that each positive result carries its limits somewhere citable rather than inline.

1. How to read this

Each entry states a limit or a closure. Labels mirror the companion: a remark labelled `O-foo` here is the discussion belonging to `rem:foo` there, where a one-sentence summary is retained in place so that no theorem stands without its scope. Numbering is independent of both other papers.

2. Scope limits of the positive results

Remark 1 (Scope and consequences)**.** Thickness is used exactly once, and sharply: the next solution would be $n_a = 2f - e$, which needs $f \geq 2e + 1$ — the thin regime. So in the regime that is open, the entire mismatch ray is now pinned on *both* sides, with no freedom beyond the order of the near-side edges. Explicitly: $N = 23$ has far 5^3, near $\{6, 9\}$; $N = 59$ far 9^5, near $\{20, 25\}$; $N = 66$ far 16^5, near $\{15^2, 25^2\}$; $N = 83$ far 11^6, near $\{30, 36\}$; $N = 167$ far 39^8, near $\{40^3, 64^3\}$. The count of interior junctions is therefore also determined — $f - 1$ on the far side and $2(f - e) - 1$ on the near side, none shared (Theorem 6(i)) — so a hypothetical thick tiling carries exactly $3f - 2e - 2$ T-junctions along this ray, each a π-vertex. The arithmetic is machine-checked (`near_side_unique`, `BaseBetaWalks.lean`).

Remark 2 (Scope, and why $m = 1$ is the rigid case)**.** Theorem 2(i) generalizes the $e = 1$ base analysis used for Theorem 1 to the infinite family $f > 2e$ — $e = 1$, $f \geq 3$; $e = 2$, $f \geq 5$; $e = 4$, $f \geq 9$; $\ldots$ — covering the primes $N = 47, 71, 107, 191, 227, 239, 359, \ldots$, and its parts (iii)–(iv), which need no thinness, deliver the $e \geq 2$ analogue of the $e = 1$ structure theorem of Remark 14 for *every* (e, f): the equal sides are b-free and the base's b-count is bounded by $j(f - e) \leq e - 1$. The step that fails for $m \geq 2$ is the very first one: there $Q \equiv em \pmod f$, so j may be *negative* and the level rises to $2em$. This is not an artifact. The genuine 44-tiling (e=1, f=2, m=2) has base walk *aaaaccca*, i.e. $(P, Q, R) = (5, 0, 3)$, which is the parameter $j = -1$, $p = 3$; and the 99-tiling (e=1, f=2, m=3) has base walk *aabbbbbbbcc*, i.e. $(2, 7, 2)$, the parameter $j = 2$, $p = 0$. Both satisfy $R \geq 1$, as the γ-trap requires.

Remark 3 (The blocking hypothesis cannot be weakened to "both ends are vertices")**.** It is worth recording what the step left open in Remark 58 may *not* appeal to. In Lemma 51 the hypothesis that the chord is blocked at both ends is what converts it into a union of whole tile edges; one cannot replace it by the weaker hypothesis that both ends are vertices of the tiling, because the tilings in this family are not edge-to-edge. The certified 99-tiling of member $(1, 2)$

Date: August 29, 2026.

settles this concretely. Its 99 tiles span 216 distinct edges, and 52 of those carry a vertex of the tiling in their relative interior — each exactly one. Moreover its 89 vertices contain 19 pairs at distance exactly a, 23 at distance exactly b and 45 at distance exactly c that span no tile edge at all. (These counts are read off the coordinates of the machine-checked certificate; in that embedding every x is rational and every y a rational multiple of $\sqrt{15}$, so the chart $(x, y/\sqrt{15})$ is an affine rescaling of the plane and decides collinearity and betweenness faithfully.)

Hence knowing that a segment has both endpoints at tiling vertices and has the length of a tile side carries *no* information about how it meets the tiling. Since V and E are vertices at distance exactly a, this is precisely the shape of the hypothesis one is tempted to use, and it is not available. Whatever forces $[V, E]$ to be a whole a-edge must consume the cascade's tile data — which tiles have been forced and where — and not merely its vertex set. The same remark explains why the tile-interior blocking of Remark 52, and not a vertex-incidence argument, is the mechanism the cascade actually runs on.

Remark 4 (A gap in the induction of Theorem 99)**.** The induction step of Theorem 99 argues that at $V_k = (kc, 0) + a\,u$ the tiles T_k and P_{k-1} present γ and β, "leaving α". That is the straight-junction identity $\pi - \gamma - \beta = \alpha$, which follows from $3\alpha + 2\beta = \pi$ and $\gamma = 2\alpha + \beta$. It is legitimate at $k = 0$, where $V_0 = a\,u$ lies on an equal side and the junction is genuinely straight. For $k \geq 1$ the point V_k is *strictly interior*: the angle around it is 2π, and

$$2\pi - \gamma - \beta \;=\; 4\alpha + 2\beta \;=\; \pi + \alpha \;\neq\; \alpha.$$

The discrepancy is exactly π. For instance at $(e, f) = (2, 5)$ one has $V_1 = (30.68, 8.23)$, interior to the target $(0, 0), (142, 0), f^3u$; the residue there is $203.07°$, not the $\alpha = 23.07°$ the step requires. The same holds at every member we computed, $(2, 3)$, $(3, 8)$, $(1, 4)$ and $(3, 7)$ among them. With the residue $\pi + \alpha$ the partner P_k is not forced, and the chain stops after T_0, P_0, T_1; that much is exactly Proposition 28.

At $e = 1$ the theorem is nevertheless true, by a different and much shorter route: the column $(0, e, 2e) = (0, 1, 2)$ has a base word of only three letters, and Proposition 28 places the first two and the last two base edges in $\{a, c\}$, which for a three-letter word is all of them; the word then carries no b-edge, contradicting $y = 1$. That argument uses no induction and no separation.

At $e \geq 2$ the word has $3e \geq 6$ letters, the four pinned positions leave $3e - 4 \geq e$ interior slots, and the b-edges fit inside them: at $(2, 5)$ the word $c\,c\,b\,b\,c\,c$ satisfies $25 + 25 + 21 + 21 + 25 + 25 = 142 = e(3f^2 - e^2)$ and survives every filter we have. So Theorem 99 is established at $e = 1$ only, and its conclusion at $e \geq 2$ — on which Corollary 100 depends — must be regarded as **open**. We have not refuted the conclusion; we refute only the proof. The missing ingredient is sharp: force the β-tile at V_k, whose flanks are $\{a, c\}$, to lay its c rather than its a along $[V_k, V_{k+1}]$. This is an instance of the crossing question of Remark 14, which Remark 45 already identifies as the obstruction, and Remark 69 already warns that a pinning argument must say against what the ends are pinned. Here the would-be blocking edge at the upper end is P_k's own c-edge, the very thing being derived.

Remark 5 (The gap is exactly one crossing, and there is no cheaper repair)**.** The step admits a sharp reduction. Let Q be the tile across T_k's b-edge $[V_k, ((k+1)c, 0)]$, whose base end lies on ∂ABC and whose end V_k is interior. Measured from the anchored base end, Q lays a, b or c along that edge:

- b: an exact match, so Q is the partner P_k — the case the induction wants;
- c: overruns V_k by $c - b = e^2$;
- a: falls short of V_k by $b - a$, leaving a run of that length still to be covered.

The third case does not close either, because $b - a$ is representable in $\langle a, b, c\rangle$ for no member whatever. Indeed if $xa + yb + zc = b - a$ then $y = 0$, since $y \geq 1$ would give $yb \geq b > b - a$;

reducing $x\,ef + z\,f^2 = f^2 - e^2 - ef$ modulo f then gives $f \mid e^2$, hence $f = 1$ by coprimality. (`gap_b_sub_a`; checked on all 29435 coprime members with $e \le 200$, $e < f < 4e+3$, with no representation in any of them.)

So both surviving branches force a tile edge past V_k, and the induction step of Theorem 99 is *equivalent* to the assertion that nothing overruns V_k. That is the one-end-anchored pinning question in its purest form: the base end cannot be overrun, because the continuation would leave the target, while V_k is interior and, as computed in Remark 104, is not blocked by any tile already forced. Repairing Theorem 99 at $e \ge 2$ is therefore neither more nor less than settling that question, and the same question is what stops the corner cascade of Remark 58 and the filler dichotomy of Section 5.6. Three routes, one wall.

Remark 6 (Counting cannot decide the crossing question)**.** Since an overrun at Y is equivalent to the tiles with a *vertex* at Y forming a π-figure, it is natural to attack it with the vertex census of Lemma 89, which counts exactly those figures. That cannot work, for a structural reason worth recording so it is not attempted again.

Write n_1 for the number of $\{\gamma,\alpha,\beta\}$ figures and n_2 for the number of $\{3\alpha, 2\beta\}$ figures, and $v_1,\dots,v_4$ for the four interior types. The census consists of one equation per angle, and its identity is obtained by subtracting the β equation from the α equation:

$$v_1 \;=\; 1 + n_2 + v_3 + 2v_4,$$

in which n_1 *cancels*. The third equation then pins $n_1 = N - 3v_1 - 2v_2 - v_3$, but supplies no bound of its own. So the census constrains $\{3\alpha, 2\beta\}$ and says nothing bounding about $\{\gamma,\alpha,\beta\}$.

That is precisely the wrong way round for us: at V_k the forced tiles are T_k, presenting γ, and P_{k-1}, presenting β, and $\{\gamma,\alpha,\beta\}$ is the only π-figure containing both. The census is therefore blind to exactly the configuration in question, and no counting argument built on it can decide the crossing.

The kernel-checked 99-tiling illustrates both points. Its vertex figures are: on the boundary, 22 of type $\{\gamma,\alpha,\beta\}$, with the corners contributing $\{\beta\}$ twice and $\{3\alpha\}$ once; in the interior, 50 of type $\{\gamma,\alpha,\beta\}$, 2 of type $\{3\alpha, 2\beta\}$, and $(v_1, v_2, v_3, v_4) = (4, 7, 1, 0)$. The identity reads $4 = 1 + 2 + 1 + 0$, exactly. And the $50 + 2 = 52$ interior π-figures agree exactly with the 52 edges carrying an interior vertex found in Remark 59 by an unrelated computation — a useful cross-check on both. Note the ratio: 50 of the 52 crossings are of the type the census cannot see.

Remark 7 (What this buys, and what it does not)**.** Corollary 116 is independent of Theorem 99 and of the base walk: it constrains the side at any member. On the close pairs with $f > 2e$ — 203 of them with $e \le 40$, the smallest being $(3, 7)$ with $N = 138$, then $(4, 9)$ with $N = 227$ and $(5, 11)$ with $N = 338$ — the side parameter is bounded by 2. Checked on all 126003 close pairs with $f > 2e$ and $e \le 1000$: $\lfloor (f-1)/e \rfloor = 2$ in every one.

On that family Corollary 101 also reduces the base to two columns. It does not reduce it to one: that step needs Theorem 99 at $e \ge 2$, which Remark 104 leaves open. So for these members the current state is two base columns and two side parameters, not one of each.

The contrast with $e = 1$ is the point. There the range is $1 \le p \le f - 1$ and grows without bound with f, which is why the corner cascade of Conjecture 60 has to be run as an induction over block lengths. Here the range is $\{1, 2\}$ for every member of an infinite family, so the side question is a fixed finite case analysis rather than an induction. This is the first sub-family in which that happens.

At $(3, 7)$, concretely: $p = 1$ gives a side of 7 a-edges and 4 c-edges, and $p = 2$ gives 14 a-edges and 1 c-edge. Excluding those two remains open; the reduction bounds the problem, it does not solve it.

Remark 8 (Why this does not exclude a member, and the trap it sets)**.** It is tempting to combine Lemma 118 with the boundary walks. On a member all of whose walks have odd edge-counts — $(3,7)$ is one, with $k = 4p+7$ on a side and 9 or 13 on the base — the boundary contributes $\sum k_i - 3$ straight figures, an even number, against the odd value $N+1 = 139$ that Lemma 118 demands. That looks like an exclusion. It is not.

S counts *every* straight figure, and a straight figure need not lie on the boundary: at an interior T-junction one tile has the vertex in the relative interior of a side, contributing a straight angle and no corner, while the tiles on the other side have corners summing to π. Such a vertex is a $\{\gamma, \alpha, \beta\}$ or $\{3\alpha, 2\beta\}$ figure and is counted in $n_1 + n_2$. Writing T for their number, $S = \sum k_i - 3 + T$, so the parity constrains only T: at $(3,7)$ every tiling has an *odd* number of interior T-junctions. Since non-edge-to-edge incidences are exactly what this problem must allow, nothing is excluded.

This is the failure mode Lemma 89 already warns of — corner counting cannot see the side structure — and the boundary walks are side structure. It is recorded here because the temptation is sharp and the arithmetic is otherwise correct.

Remark 9 (What this does and does not extend)**.** Theorem 125 uses $e^2 + ef < f^2$ at exactly one step: that Y's edge from P towards J, being of length at most $|PJ| = a$, is therefore *equal* to a. That inference needs a to be the shortest side, and it fails without it: at $(2,3)$, $(3,4)$, $(4,5)$, $(5,6)$ the tiles are $(6,5,9)$, $(12,7,16)$, $(20,9,25)$, $(30,11,36)$, each with $b < a$, so a b-edge also fits inside $|PJ|$; since b is a flank of α, the conclusion "Y does not present α" no longer follows.

Proposition 126 removes the hypothesis from Theorem 125, and with it from Corollary 130: the bound $pe + 2 \le f$ holds at every coprime member. It does *not* remove it from Theorem 167, whose third cross product is a positive multiple of $b^2 - a^2$ and changes sign at the threshold; there the hypothesis is genuinely part of the statement.

Remark 10 (Scope, and the status of this theorem)**.** The family $e = f - 1$ was outside the reach of §5.8 until Proposition 126 removed the golden-ratio hypothesis: $f/(f-1) < \varphi$ for every $f \ge 3$, so every member of it has $b < a$. Its tile counts are $N = 2f^2 + 2f - 1$, giving $11, 23, 39, 59, 83, 111, 143, 179, 219, 263, 311, \ldots$, of which 23 and 39 were previously settled by exhaustive search, 11 by Theorem 1, and 59 by the frontier results for $N \le 80$. Among the remainder the base-β candidates, that is the primes congruent to 11 modulo 12, are

$$83,\ 179,\ 263,\ 311,\ 419,\ 479,\ 683,\ 839,\ 1103,\ 1511,\ 2111,\ 2243,\ 2663,\ 2963,\ 3119$$

for $f \le 39$. Whether the family contains infinitely many such primes is a Bunyakovsky-type question and is not claimed here.

The theorem discharges Hypothesis 29 at these members; it does not by itself exclude the corresponding integers, which requires the remaining branches.

We record the label honestly. The chain is PROVED, not VERIFIED. Lean carries the arithmetic (`ApexRigidity.b_gt_f`, `.eb_gt_a`, `.e_ge_two_of_b_lt_a`, `.side_p_bound`), the area accounting (`.area_above_chord`, `.middle_fraction`) and the vertex budget of Corollary 123 in both branches (`SecondEdge.noTP_budget`, `.noTP_remainders`, `.T2_not_alpha_at_P`). What is not formalized is the planar geometry underneath: that T_1 presents γ at P and that T_2's chord trace ends there, which are the hypotheses those statements take. The non-representability of $a - b$ was checked symbolically, with $(f-2)(2f-1) - (a-b) = (f-1)^2$, and by exhaustive enumeration for $3 \le f \le 39$. Two adversarial passes have been made, both within the session that produced the proof. The first found a genuine defect — Corollary 130 was stated under a hypothesis that fails at $e = f - 1$, the very members it was invoked for — now repaired by restating that corollary under $\gcd(e,f) = 1$. The second checked the side equation, integrality of p, the case $p \geq 2$ including $f = 2$, that J is interior to the side so that the tile Y exists at all, the edge counts, absence of circularity, and that Theorem 122 and Corollary 123 use nothing beyond family-wide facts; it found no further defect.

The identification with Hypothesis 29 is the one discussed in Remark 30 and is not settled; the theorem is stated in terms of p for that reason. Theorem 128 is therefore unconditional as a statement about p, and inherits that identification when read as a statement about the hypothesis.

A second point in its favour: since $N = m^2(3f^2 - e^2)$, a prime N forces $m = 1$, so at each of the fifteen candidates the branch has no other scale to consider.

Neither pass is what Rule 6 asks for, namely a review in a fresh session with no memory of the derivation, and that remains outstanding. Each of the fifteen candidate integers above has, we note, a unique base-β member, so discharging the hypothesis at $(f-1, f)$ discharges it for that integer's only member of this branch.

Remark 11 (What this settles, and exactly where it stops)**.** Corollary 130 is a second and independent proof of Theorem 163 on the tight subfamily, by a route that uses no chord bound and no height identity — only the apex identities and the boundary at J. It also completes the first bullet of Remark 166: there the $a\,c$ ending was excluded only in the figure $\{\gamma, \alpha, \beta\}$ and only with the α adjacent to C, whereas Theorem 125 excludes it in *both* straight figures and for every p. Hence at $(3, 7)$ the side word of an equal side must end $c\,c$; the two cases $X = \beta$ and $X = \gamma$ of Remark 166 both persist, but now only inside that ending.

It does not reach $p = 1$ at any new member, and the reason is arithmetic rather than incidental. Killing $p = 1$ by this bound needs $e \geq f - 1$; with $e^2 + ef < f^2$ that forces $f = e + 1$ and then $e^2 < e+1$, so $e = 1$ and $f = 2$ — the member already closed by Theorem 1 (`scope_limitation`). The hypothesis $e^2 + ef < f^2$ says f/e exceeds the golden ratio, and $e \geq f - 1$ says it is close to 1; the two cannot hold together beyond $(1, 2)$.

The obstruction to iterating is equally precise. Corollary 123 needed *two* tiles presenting γ at P, and the second, T_2, exists only because the apex figure is 3α. At the analogous point one level down — the far end of the next c-tile's a-edge, which does lie exactly on the next chord, by the same identities — only one γ is forced. Supplying a second is what an induction would require, and we do not have it.

Remark 12 (What this buys, and what it does not)**.** This is a reformulation, not a new obstruction: the question "is the inflation rigid" and the question "is $p = 0$" are the same question, and at $(1, 3)$ Theorem 53 already answers it. Reformulations earn their place only by new attack surface, and here there is one, namely size. The inflation carries f^2 tiles against the target's $N = 3f^2 - e^2$:

(e, f)	f^2	N
$(1, 3)$	9	26
$(2, 5)$	25	71
$(3, 7)$	49	138
$(4, 9)$	81	227

uniformly about 35%. At $(3, 7)$ that is a 49-tile question in place of a 138-tile one, with the same answer for the west side. Proposition 140 carries no unproved geometric hypothesis: the γ-corner, the only one that can split, is not used.

Remark 13 (What this settles and what it does not)**.** Remark 47 draws exactly this inference — "the complete west block is the tile scaled by f, so its far side ... is subdivided into exactly f c-edges" — and asserts the middle step rather than proving it. Corollary 144 proves it at the three members where the computation has been run, $(4, 9)$ included.

It does not decide p at $(3,7)$. The inflated tile is the west corner block only if that block is complete, which is the west half of Hypothesis 29 itself; the corollary is conditional on exactly the hypothesis it serves. What has been removed is an unproved step *inside* that implication, not the implication's hypothesis.

At $(1,3)$ the theorem also reproves Theorem 53 by an independent route, on 9 tiles rather than 26, which is the regression check that the reformulation of §5.9 recovers a known case before being trusted elsewhere.

Remark 14 (What this buys)**.** Two things, on different fronts.

On the inflation, The junction analysis of §5.9 (machine-checked as `junction_three`) leaves three completions at each junction of the $k = 2$ word $c\,a^f c$, hence 3^{f-1} patterns a priori. Proposition 149 leaves $f+1$, one for each value of j: the branching is linear in f, not exponential. That is consistent with the measured search depth, which grows like $2.65f$ rather than with f^2.

On the thin family, nothing in the proof used the inflation — only that an a-edge's two ends carry β and γ. So the proposition applies verbatim to an a-run on an *equal side*, and in particular to the runs enumerated in the second gap of Conjecture 60. That enumeration is over the double-c walks $(fk, 0, f-k)$ with $1 \le k \le f-3$; the number of such words grows quickly with f $(1, 7, 42, 263, 1823, \ldots$ for $f = 4, \ldots, 8)$, so it is not a finite check, but the orientation branching within each word is now linear rather than exponential. This is the first tool bearing on that gap which is uniform in f; the cascade argument as written does not use it.

Remark 15 (Scope, and why this is not a proof of Theorem 143)**.** Two limitations, both essential.

First, $\gcd(e,f) = 1$ forbids e and f both even, so "$e \not\equiv f$" is exactly "one of them is even". The proposition is therefore *silent* on the both-odd members — which include $(1,3)$ and $(3,7)$, i.e. every member the crux actually needs. It fires at $(2,5)$ and $(4,9)$, where $3f^2-(4f-e)$ equals 57 and 211, and those are precisely the two members the search found cheap. That correlation is the content of the proposition: it explains the observed cost profile, it does not extend the theorem. The $(4,9)$ search has since returned `EXHAUSTED_NO_TILING` in 5 161 190 nodes, which agrees with the parity verdict there and is strictly stronger, the search assuming no edge-to-edge condition.

Second, edge-to-edge is an extra hypothesis, and it is not a mild one here. The main paper does not assume it, and the T-junction whose exclusion is the whole difficulty of §5.8 is exactly an edge-to-edge failure; assuming it would be assuming a neighbourhood of what is to be proved. This is the same wall that Remark 18 hits: the b-edge pairing argument there fails precisely because a longer edge may straddle a b-edge instead of abutting it, and both genuine tilings exhibit such straddles. Proposition 146 is what survives once straddles are hypothesised away, which is why it proves less than it appears to.

So the uniform proof of Theorem 143 remains open. What Proposition 146 shows is that any such proof must use more than counting on the both-odd members, since the counting invariant is identically satisfied there. (Machine-checked, `parity_kills_25`, `parity_kills_49`, `parity_silent_witnesses`.)

One caveat should be entered in the hypothesis's favour. The edge multiset of the two standard inflations was computed from the certified tilings in `code/engine/inflation/`: at $(1,3)$ every edge has multiplicity 1 or 2 with 9 of the former and 9 of the latter, and at $(3,7)$ with 21 and 63; no edge has multiplicity above 2, and $2 \cdot 9 + 9 = 27 = 3 \cdot 3^2$, $2 \cdot 63 + 21 = 147 = 3 \cdot 7^2$. So the inflation that does exist *is* edge-to-edge, and the boundary count is $3f$ on the nose. That is a positive control sensitive to exactly the quantity the proof above manipulates — a mis-stated B would break it — and it is recorded because the vertex-census parity of Remark 119 was retracted after passing a control that could not see its defect. What the control cannot do is speak for the $p = 1$ boundary, since no such tiling exists to measure.

Remark 16 (The last junction, and what is still missing)**.** At J_{2f} the two boundary neighbours are the a-tile A_{2f}, presenting γ with a b-ray inward, and the apex c-tile, presenting β with an a-ray inward; the filler's rays are b and c. Since neither b nor c equals a, J_{2f} *cannot be edge-to-edge*, and each placement leaves a residual run: $c - a$, or $b - a$ together with $c - b = e^2$. All three are gaps of $\langle a, b, c\rangle$ — $b - a$ by `gap_b_sub_a`, e^2 by `gap_e_squared`, and $c - a$ by `gap_c_sub_a_tight`, which holds precisely for $e \geq 2$ since $c - a = 2a$ at $(1, 3)$. Moreover $e^2 < a$, so no tile edge fits inside that run at all. Hence the covering *must* overrun a filler vertex.

Each placement therefore forces the covering to overrun a filler vertex. That alone is not an exclusion, and the exclusion comes instead from a chord bound, which we now state.

Remark 17 (The audit of Theorem 163 against the apex rigidity)**.** Theorem 163 predates §5.8, and the two overlap. Reconciling them shows the earlier enumeration was doing more work than necessary, and we record this rather than leave two arguments whose relation is unclear.

Remark 161 has the α-filler standing between the apex c-tile and the a-tile, so it lays one of its flanks — a b or a c — along the apex tile's a-ray. Both exceed $a = |JP|$, so P would lie *interior* to that edge, contributing π at P. But Theorem 122 puts T_1 at γ there, and T_2's trace on the chord ends at P, so T_2 contributes at least γ as well. The total is at least $2\gamma + \pi = 2\pi + \alpha > 2\pi$.

So that configuration does not occur at all, and the placements enumerated in the proof of Theorem 163 are placements of a tile that cannot be there. The conclusion is unharmed — indeed Corollary 130 re-derives it, on a wider range of members and without any chord estimate — and Theorem 165's conclusion, that the adjacent tile lays an a-edge, is likewise immediate from the same count. What the audit removes is the impression that the tight subfamily needs the height identity: it does not.

Remark 18 (Scope, and where $e \geq 2$ is easier)**.** Theorem 163 closes $p = 2$ on the whole subfamily $f = 2e+1$, hence at $(1, 3)$, $(2, 5)$, $(3, 7)$, $(4, 9)$, $(5, 11)$ and onward; verified independently by exact enumeration of the six placements for every member with $e \leq 400$. Combined with Corollary 116 it leaves $p = 1$ as the only obstruction to Hypothesis 29 on that subfamily.

It does not extend beyond $f = 2e + 1$: the height identity used above holds exactly there.

Two by-products. First, $c - a$ is a gap of $\langle a, b, c\rangle$ precisely for $e \geq 2$, failing at $(1, 3)$ where $c - a = 2a$; this is the first point in the problem where $e \geq 2$ is easier than $e = 1$. Second, at $(3, 7)$ the base walk $(0, e, 2e)$ carries no a-edge, so the corner tile lays c on the base and a on the side, forcing $p \geq 1$ on both equal sides; with $p = 2$ excluded, Hypothesis 29 at $(3, 7)$ therefore requires the base walk (f, e, e).

Remark 19 (What the two no-go results mean)**.** Proposition 30 shows that the local combinatorics of the side is almost unconstrained: the automaton forbids one adjacency and leaves 10 560 configurations. Proposition 31 shows that adjoining the census adds nothing, because the two counting equations are dependent. This is the third independent confirmation of the warning in Lemma 89 that corner counting cannot see p, and it is also the exact reason the retraction described in Remark 119 was possible: a counting argument at $(3, 7)$ has one relation to spend and it is spent.

The conclusion is directional, not merely negative. Any mechanism that closes $p = 1$ must use metric information — positions, not multiplicities. Proposition 33 is the first such fact.

Remark 20 (The wedge-filler hypothesis, and where it fails)**.** Several steps of the corner chain fill a wedge of angle exactly α with a single α-tile, justified in Lemma 70 by "α is the minimal angle". That is not the load-bearing fact. What the step needs is that no two smaller angles fit, that is $2\beta > \alpha$, which with $3\alpha + 2\beta = \pi$ reads

$$\alpha < \tfrac{\pi}{4}.$$

The two conditions are different, and both directions of the difference occur. At $(e,f) = (3,4)$ one has $b < a$, so β is the minimal angle, yet $\alpha < \pi/4$ still holds and the step survives.

The criterion is exact and integral. Since $b^2 + c^2 - a^2 = (f^2 - e^2)(2f^2 - e^2)$ and $2bc = 2f^2(f^2 - e^2)$,

$$\cos\alpha = \frac{2f^2 - e^2}{2f^2} = 1 - \frac{e^2}{2f^2},$$

which at $e = 1$ is the familiar $(2f^2 - 1)/(2f^2)$. So $\alpha < \pi/4$ iff $2f^2 - e^2 > \sqrt{2}\, f^2$, and squaring (both sides positive, as $f > e$) gives

$$e^4 - 4e^2 f^2 + 2f^4 > 0.$$

The separation condition $f^2 > 2ef + e^2$ implies it (`separated_wedge_ok`): from $f^2 - e^2 > 2ef$ one gets $f^4 + e^4 > 6e^2 f^2$ on squaring, and the claim follows with room to spare. So the step is safe at every separated member, and every failure is a close pair. Among $e \le 11$, $f \le 19$ the failures are

$$(4,5),\ (5,6),\ (6,7),\ (7,8),\ (7,9),\ (8,9),\ (9,10),\ (9,11),\ (10,11),\ (10,13),\ (11,12),\ (11,13),$$

including $(5,6)$, which is $N = 83$. No claim in either paper is affected: Lemma 70 is invoked only at $e = 1$, where $a = f < f^2 - 1 = b$ and $\alpha < \pi/4$ both hold, and Proposition 41 lists four ingredients not including it. What the criterion supplies is a precise location for the obstruction: close pairs differ from separated members geometrically, not merely in having weaker arithmetic filters, and an argument aimed at them must either avoid the wedge-filler step or treat $e^4 - 4e^2 f^2 + 2f^4 \le 0$ separately.

Two arithmetic strengthenings do *not* help there, which is why the bound must be interior. Reducing the base equation modulo e gives $y + z \equiv 0 \pmod{e}$ by coprimality; and the γ-injection is exactly a bipartite matching of the a- and b-edges to the junctions, whose sharp criterion is Hall's condition rather than the counting bound. Neither removes any of the 54 columns that survive the three unconditional filters over $e \le 9$, $f \le 15$: the first is implied by the base equation, and the second collapses to the γ-trap $z \ge 1$, since with $z \ge 1$ a matching always exists.

Remark 21 (What reach buys, and what it does not)**.** The four kills at reach R empty the window exactly for $f \le R + 2$: reach 3 gave $f \le 5$, reach 4 gives $f \le 6$, and each further row would give one further member. Theorem 80 raises the reach by one and no more, because the clearance bound it uses is the fixed constant $T/b > 2$. A proof for all f therefore requires killing straddlers at *every* row, not at the first three. The natural route is that a ladder descends into strips the chain has already forced, where every line of its direction is a shared b-edge chain broken at the strip junctions on both sides at once — unstaggered — while a ladder is staggered by $c - b = 1$; a ladder in forced territory would have to run along such a line or cross tile interiors, and both are impossible. What is missing is the bookkeeping: that the region a ladder descends through, roughly one half-column east per strip, lies inside the forced staircase at the moment the fork is being decided.

Remark 22 (What this does to the induction)**.** The collar of Theorem 182 runs on unit columns, so it is unavailable at $e \ge 2$; only the wide columns of Proposition 195 remain, and they advance m in steps of f. Consequently the clean law "tileable iff $m \ne 1$" proved for $(1,2)$ need not persist: the spectrum may genuinely depend on e. For $(2,3)$, where $f = 3$, the subdivision closure and the wide-column law together would give every $m \ge 4$ and every even m from a single seed Δ_2, leaving exactly $m = 3$ unreached from below; so the decisive experiment for the whole $e \geq 2$ story is the realizability of Δ_2 at $(2, 3)$, i.e. $N = 92$. That search is running; we record the reduction rather than guess its outcome.

Remark 23 (The chord programme is exhausted)**.** It is natural to hope that the family of chords does collectively what no single chord does: there are 37 admissible heights $\rho = n/49$ with $n = 3i + 7j$, each carrying its own non-integral area $138\rho^2$, and a tile lying below the chord at J lies above the chord at some lower junction. We record that this hope is also unfounded, so that the reader need not pursue it.

For a tile T and a chord C write $\varphi_T(C) \in [0, 1]$ for the fraction of T's area lying above C; thus T straddles C exactly when $\varphi_T(C) \in (0, 1)$. The area statement at height ρ is

$$\sum_T \varphi_T(C_\rho) \;=\; 138\rho^2,$$

and this is nothing but additivity of area over the tiling: the left-hand side *is* the area above C_ρ, decomposed tile by tile. It therefore holds identically, for every tiling of the target, and the 37 admissible heights supply 37 instantiations of one identity rather than 37 constraints.

The only information extractable from the family is the integrality bit — where $138\rho^2 \notin \mathbb{Z}$, some $\varphi_T(C_\rho)$ lies strictly between 0 and 1 — and that is precisely Proposition 33. The nesting adds nothing further: the set of heights straddled by a given tile is an interval, and 138 tiles cover 37 heights with room to spare.

This is the same degeneracy as Proposition 31, in a metric rather than a combinatorial register: a single global identity, instantiated many times, cannot separate tilings. Taken together with Remark 37 it closes the chord programme. What the chord method yields at $(3, 7)$ is exactly Propositions 33 and 36, and closing $p = 1$ requires a mechanism that is neither counting nor chord-area. We do not have one, and we prefer to say so.

Remark 24 (Where the chain stops being free)**.** Lemma 44 iterates only as far as the same two ingredients survive. The fill of $\alpha + \beta$ is unique at every member, so the two tiles are always an α and a β; what decides whether the next base edge is confined to $\{a, c\}$ is their angular order, and that is settled by whether the α-tile is the partner across the previous tile's whole b-edge. At the corner this is unconditional, because the b-chord runs from a base vertex to a side vertex. At the k-th step the chord's upper end is interior, and pinning it is precisely the crossing question of Remark 14. Both remaining cases, the residual configurations at $e = 1$ and the exclusion of $1 \leq p \leq (f - 1)/e$ at $e \geq 2$, reduce to that single question.

That question is not answered by the hypothesis $m = 1$ alone. Instrumenting the certified exhaustions to record every partial configuration they build shows the overshoot occurring freely: at $(1, 2)$ it appears in 31 of the 52 recorded frontiers and at $(1, 3)$ in 1128, the first already at depth 3 with three tiles placed, a c-edge from $(2, 0)$ to $\left(\frac{11}{2}, \frac{\sqrt{15}}{2}\right)$ covering a b-edge from $(2, 0)$ and overshooting it by $c - b = 1$. Every such configuration is eventually refuted, but not locally: the partner pinning cannot be established by excluding the overshoot near the chord.

Remark 25 (The thick program)**.** The three verified facts above change the character of the $e \geq 2$ case. The chain machinery ports with strictly stronger tools: the wall climb and the first-run chirality are member-independent (their proofs use only the angle identities and the boundary overshoot $c - b = e^2 > 0$), and Lemma 93 removes the c-chord fork entirely, so the thick chain branches strictly less than the thin one. What remains is the finite bookkeeping: the base letters run through a trichotomy rather than a free word, the side violations are quantized to $p \in f\mathbb{N}$, and the reach analysis of the thin case applies verbatim where the blocks are short. The $(2, 3)$, $(2, 5)$ and $(3, 4)$ members are certified by full exhaustions (19,677, 553,417 and 825,724 nodes; the second is the prime $N = 71 \equiv 11 \pmod{12}$, the third a close pair). The close pair refutes deeper ($8.3f$ against $5.7f$ and $5.2f$), consistent with its additional base decompositions.

Remark 26 (Why this is the right direction)**.** At the last junction, $n = 7$ and the area above the chord is $138/49 = 2.816\ldots$ tiles, while the chord has length $414/7 = 59.14\ldots$ and a tile meets a line in a segment of length at most $c = 49$; so at least two tiles meet that chord, and at least one straddles it. This is exactly the region below the chord that the enumeration of Theorem 163 does not reach — see the discussion of the $c\,c$ ending in Remark 166 — and it is the reason the surviving configurations survive: not that the chord bound fails there, but that it was never applied there. Whether the straddling constraint can be sharpened into a contradiction is open.

Remark 27 (What the decomposition gives, and the rung that remains)**.** Proposition 36 is the first statement that constrains the configuration *below* the chord at J, the region Remark 166 identifies as untouched. It reduces the possibilities there to a finite list: a flush total in $\{0, 21, 40, 42, 49\}$ together with a straddling remainder of $59.142\ldots$, $38.142\ldots$, $19.142\ldots$, $17.142\ldots$ or $10.142\ldots$, distributed over at least one and at most a bounded number of straddling tiles.

What it does not give is rigidity: the cross-section of a straddling tile varies continuously with its position, so no individual value is forced. The natural hope is that the γ-grading of §5.2 restores discreteness — a straddler's edges have directions that are integer multiples of γ modulo π, so one might expect its cross-section to be drawn from a countable set fixed by its label. *That hope is unfounded*, and we record why, since it is the first thing a reader will try.

The grading constrains directions, not positions. Tile vertices lie in the $\mathbb{Z}$-module generated by the admissible edge vectors, and for a horizontal chord what matters is the projection of that module to the vertical axis. Since $\gamma = (\pi + \alpha)/2$ with α/π irrational, that projection is generated by the values $\ell \sin(k\gamma)$ for $\ell \in \{21, 40, 49\}$ and k in the label window, and these are rationally independent: the subgroup they generate is dense in $\mathbb{R}$. So no cross-section is forced, and the chord decomposition cannot by itself close $p = 1$.

What the decomposition does establish is that the obstruction is located strictly *below* the chord. Above it there is only $138/49 = 2.816\ldots$ of a tile's area, so that region is enumerable up to boundedly many tiles — which is precisely the enumeration Theorem 163 performs. The surviving configurations of Remark 166 place the decisive tile below the chord, where no area bound confines it. Closing $p = 1$ therefore requires a constraint that reaches downward, and we do not have one.

Remark 28 (The residue at $f \geq 6$)**.** For $f \geq 6$ the surviving words form an explicit thin set, symmetric under reversal; at $f = 6$ it is a single class: b at position 5, c at position 7 ($a\,a\,a\,a\,b\,a\,c\,a$), read from a corner whose block is at least 3. The blocking structure is always the same: the chain's column-3 line needs a row-3 brick, the row-3 fork's A-branch lays its south cover on the row-2 floor, and the floor is edged only as far as the staircase has reached — which the b itself interrupts. Closing the row-3 fork is the entire remaining content of Conjecture 60 at $e = 1$.

3. Routes closed, with their failure points

The following approaches to this branch are closed, and are recorded so that they are not retried.

(1) **The whole class of linear direction invariants.** Weighting a directed edge by a function of its direction alone, antisymmetric under $\theta \mapsto \theta + \pi$, yields a one-parameter family of identities; every member is a specialisation of the master identity of [1, Rem. 4.9]. The associated ideal-membership condition is satisfied identically for every root of unity and

every (e, f), and the optimal ℓ^1 bound the family can produce is $m(3f - 2e)(f + e)/f$, which is smaller than N by a factor at least $11/6$ and growing like mf. So no choice of weight, kernel, moment ladder or duality can exclude a candidate. Both statements are machine-checked in `lean/LambdaFactor.lean`.

(2) **Closing $e = 1$ by proving $R \geq 2$ on every side.** False at the base: the walk with $R_{\text{base}} = 2$ is exactly the one the corner parallelogram kills, and the survivor has $R_{\text{base}} = 1$. On the base, "$R \geq 2$" restates the remaining problem rather than reducing it. It does hold on the two equal sides, and is proved there.

(3) **Boundary tile counts.** The number of tiles meeting ∂ABC grows like the number of boundary edges, and on the genuine certificates it is 28 of 44 and 44 of 99 — a falling fraction. Comparing it against N therefore never yields a contradiction.

(4) **The far side of the mismatch ray**, under the hypothesis that the ray is a complete wall: that side is the tile scaled by f, and is tileable. No obstruction can come from there.

(5) **Local combinatorics of an equal side, with or without the census**, for $p = 1$ at $(3, 7)$. The junction automaton forbids a single adjacency and leaves 10 560 configurations (Proposition 30); adjoining the corner census adds nothing, its two counting equations being dependent (Proposition 31). Together with Lemma 89 this is the third confirmation that corner counting cannot see p.

(6) **The chord/area programme**, beyond Propositions 33 and 36. Two continuations both fail: the γ-grading does not discretize straddle positions, because it constrains directions and the vertical projection of the vertex module is dense (Remark 37); and the family of 37 admissible chord heights supplies 37 instantiations of one identity — additivity of area — rather than 37 constraints (Remark 38). Closing $p = 1$ needs a mechanism that is neither counting nor chord-area.

(7) **Boundary-word invariants (Conway–Lagarias tiling groups)** [?]. These are the natural non-abelian strengthening of the direction invariants of item 1, and are known to be strictly stronger than colouring and weighting arguments — which is exactly the class item 1 kills here. They nevertheless do not apply, for two independent reasons. First, the method is formulated for regions assembled from cells of a regular lattice, and the vertices of a tiling of our target lie in no lattice: the edge vectors have lengths in $\{21, 40, 49\}$ and directions $k\gamma$ with $\gamma = (\pi + \alpha)/2$ and α/π irrational, so the vertical projection of the $\mathbb{Z}$-module they generate is dense in $\mathbb{R}$. Second, tilings here need not be edge-to-edge, so the boundary word of a region is not determined by its combinatorics.

4. Two mechanisms that provably cannot close $p = 1$

Remark 166 ends by saying that closing $p = 1$ "requires a further mechanism". That is a hedge, and it can be replaced by theorems. This subsection records two natural mechanisms and proves that neither can work, and one arithmetic fact that survives and points the other way. The member is $(e, f) = (3, 7)$ throughout: tile $(a, b, c) = (21, 40, 49)$, target $(343, 343, 414)$, $N = 138$.

Fix an equal side with $p = 1$. Its word is a permutation of a^7c^4 ending in c ($7 \cdot 21 + 4 \cdot 49 = 343$), so it has 11 edges and 10 interior junctions. Each edge is laid by a tile, and the two angles of that tile at the ends of the edge are the two angles *not* opposite it: an a-edge is flanked by β and γ, a c-edge by α and β. Call the choice of which flank sits at the lower end the *orientation* of the edge.

Lemma 29 (Endpoint angles). [PROVED] *The angle at the top of the last edge is α, and the angle at the bottom of the first edge is β.*

Proof. By Lemma 89 the apex fills uniquely as $\{3\alpha\}$ and each base corner uniquely as $\{\beta\}$. The last edge of the side is a c-edge, whose flanking angles are α and β; the tile at the apex presents α, so the top angle is α. The single tile at the base corner presents β, so the bottom angle of the first edge is β. □

Proposition 30 (The junction automaton is consistent, and hugely so)**.** [PROVED] *Regard the side as a transfer system: a state is the angle at the top of the current edge, and a transition is admissible when the pair (top angle of one edge, bottom angle of the next) extends to a straight figure at their junction. Then the only inadmissible pair is (γ, γ), which can occur only at a junction between two a-edges. Consequently, over all 120 words a^7c^4 ending in c and all 2^{11} orientations, exactly $10\,560$ configurations satisfy every junction constraint together with Lemma 29.*

Proof. A junction of an equal side carries a straight figure, so by Lemma 89 it is $\{\gamma, \alpha, \beta\}$ or $\{3\alpha, 2\beta\}$. The figure $\{\gamma, \alpha, \beta\}$ has one angle of each kind, so it admits a pair of side angles precisely when the two are distinct. The figure $\{3\alpha, 2\beta\}$ contains no γ, so it admits precisely the pairs drawn from $\{\alpha, \beta\}$, all four of which leave an admissible remainder (3α, $2\alpha + \beta$, $2\alpha + \beta$, $\alpha + 2\beta$ respectively). A pair is therefore inadmissible exactly when it is equal and involves γ; and γ occurs only as a flank of an a-edge. The count is a finite check over $120 \cdot 2^{11}$ cases. □

Proposition 31 (The census contributes one relation, and it is already spent)**.** [PROVED] *Let x be the number of junctions of the side carrying $\{\gamma, \alpha, \beta\}$. Then $x \geq 7$, and the α- and β-counts along the side are linearly dependent: they sum to the identity $13 = 13$. Both $x = 7$ and $x = 10$ occur in solutions, so the census does not determine x.*

Proof. The 7 a-edges contribute one β and one γ each, the 4 c-edges one α and one β each: 7γ, 11β, 4α in all. By Lemma 29 the apex absorbs one α and the base corner one β, leaving 7γ, 10β, 3α distributed over the 20 side-tile slots at the 10 junctions. A $\{3\alpha, 2\beta\}$ junction contains no γ and a $\{\gamma, \alpha, \beta\}$ junction exactly one, so the 7 γ's occupy 7 distinct junctions of the latter type, whence $x \geq 7$.

Write u for the number of junctions whose γ lies on a side tile and whose other side slot is β, and (a_i, b_i) with $a_i + b_i = 2$ for the side slots at the $10 - x$ junctions of type $\{3\alpha, 2\beta\}$. Counting α and β separately,

$$(7 - u) + (x - 7) + \textstyle\sum_i a_i \;=\; 3, \qquad u + (x - 7) + \textstyle\sum_i b_i \;=\; 10.$$

Adding and using $\sum_i (a_i + b_i) = 2(10 - x)$ gives $7 + 2x - 14 + 20 - 2x = 13$, an identity. The two equations are therefore dependent. Taking $x = 10$, $u = 7$ satisfies both, as does $x = 7$, $u = 4 + \sum_i a_i$ for any $\sum_i a_i \leq 3$. □

Remark 32 (What the two no-go results mean)**.** Recorded in [**?**]. Full discussion in [**?**].

Proposition 33 (Every junction chord is straddled)**.** [PROVED] *Let J be a junction of an equal side at distance d from the apex, and let C be the chord of the target through J parallel to the base. Then some tile of the tiling meets both open sides of C.*

Proof. Since d is a sum of edge lengths 21 and 49, we have $d = 21i + 49j$ and the height fraction is $\rho = d/343 = (3i + 7j)/49$. Write $n = 3i + 7j$, so $1 \leq n \leq 48$ for a junction strictly between the apex and the base. The chord cuts off a triangle similar to the target with ratio ρ, whose area in units of the tile area is

$$138\rho^2 \;=\; \frac{138\, n^2}{2401}.$$

Now $2401 = 7^4$ and $138 = 2 \cdot 3 \cdot 23$, so $\gcd(138, 2401) = 1$ and the quantity is an integer if and only if $7^4 \mid n^2$, i.e. $49 \mid n$ — impossible for $1 \leq n \leq 48$. Hence the area above C is not an integral number of tiles. If no tile met both open sides of C, every tile would lie weakly on one side and the area above C would be a sum of whole tile areas, a contradiction. □

Proposition 34 (The general form)**.** [PROVED] *Let (e, f) be any member with $\gcd(e, f) = 1$. Then the area above the chord through any junction strictly between the apex and the base is never an integral number of tile areas, so some tile meets both open sides of that chord.*

Proof. A junction at distance $d = ia + jc = f(ie + jf)$ from the apex has height fraction $\rho = m/f^2$ with $m = ie + jf$ and $1 \le m < f^2$, so the area above its chord is Nm^2/f^4 tile areas. Since $\gcd(e, f) = 1$ we have $\gcd(N, f) = \gcd(3f^2 - e^2, f) = \gcd(e^2, f) = 1$, so $f^4 \mid Nm^2$ forces $f^4 \mid m^2$, hence $f^2 \mid m$ prime by prime, which contradicts $m < f^2$. The rest is as in Proposition 33 (`ChordDecomp.area_never_integral`; the coprimality is automatic on the tight subfamily, `ChordDecomp.coprime_tight`). Checked by exhaustive enumeration over all members with $e \le 14$, $f \le 31$ and all admissible junctions: 80 299 cases, no exception. □

Remark 35 (Why this is the right direction)**.** Recorded in [**?**]. Full discussion in [**?**].

Proposition 36 (Chord decomposition at the last junction)**.** [PROVED] *Let C be the chord of the target through the last junction J of an equal side at $(3, 7)$, so $|C| = 414/7$. Let $\mathcal{U}$ be the set of tiles whose interiors meet the open half-plane above C. Each $T \in \mathcal{U}$ either* straddles *C (its interior meets both open sides) or is* flush from above *(it lies weakly above the line of C). Then:*

(a) *a flush tile meets the line of C in a full tile edge, of length 21, 40 or 49;*
(b) *the traces on C of the tiles of $\mathcal{U}$ have pairwise disjoint interiors and cover C, so their lengths sum to $414/7$;*
(c) *at most two tiles of $\mathcal{U}$ are flush, and the multiset of flush lengths is one of $\varnothing, \{21\}, \{40\}, \{49\}, \{21, 21\}$;*
(d) *at least one tile straddles, and if none is flush then at least two straddle.*

Proof. (a) For a tile lying weakly above the line of C, that line is a supporting line, so the intersection is a face of the tile — a vertex or an edge (`contact_is_edge`). A trace of positive length excludes the vertex case, and the edge lengths are $a = 21$, $b = 40$, $c = 49$.

(b) The closed region above C is covered by the tiles, and a point of C interior to the target is a limit of points of that region, hence lies in a tile of $\mathcal{U}$, which is closed; so the traces cover C. If two tiles of $\mathcal{U}$ had traces overlapping in a segment, both would occupy area immediately above that segment, contradicting disjointness of interiors. Additivity of length gives the sum.

(c) The flush lengths are integers drawn from $\{21, 40, 49\}$ and, by (b), sum to at most $|C| = 414/7 = 59.142\ldots$. The only multisets meeting that bound are the five listed: already $21 + 40 = 61 > |C|$, and $21 + 21 + 21 = 63 > |C|$.

(d) By the computation in Proposition 33 the area above C is $138/49$ tiles, not an integer, so a straddler exists. If no tile of $\mathcal{U}$ is flush, the straddling traces alone sum to $|C| = 59.142\ldots$; a segment contained in a tile has length at most the tile's diameter, which for a triangle is its longest side $c = 49$. Hence at least two straddlers. □

Remark 37 (What the decomposition gives, and the rung that remains)**.** Recorded in [**?**]. Full discussion in [**?**].

Remark 38 (The chord programme is exhausted)**.** Recorded in [**?**]. Full discussion in [**?**].

The clause "that admits a solution" in Proposition 177 is not decided by (iii): the bound there ignores the residue class of x, and over-permits. At $(e, f) = (8, 17)$ it allows $k = 1$, while $8x + 17z = 47$ with $x \equiv 8 \pmod{17}$ has no solution with $x, z \ge 1$, so no column exists; the same happens at 551 of the 1558 close pairs with $e \le 60$. Restoring the residue class makes the test exact.

Proof. (i) By Proposition 177(i),(ii) a column satisfies $xe + zf = 2ef - kb$ and $x \equiv ke \pmod f$; conversely, for any x in that class f divides $2ef - kb - xe$, because $xe \equiv ke^2$ and $2ef - kb \equiv ke^2 \pmod f$. So the column exists precisely when some $x \geq 1$ in the class has $z = (2ef - kb - xe)/f \geq 1$, that is $xe + f + kb \leq 2ef$. The left side increases with x, so it suffices to test $x = x_k$, and then $x = x_k$ by Proposition 177(iv). Writing $x_k = ke - m_k f$, the identity

$$(ke - mf)e + f + k(f^2 - e^2) - 2ef \;=\; f\big(kf - (m+2)e + 1\big)$$

turns that inequality, after dividing by $f > 0$, into $kf \leq (m_k + 2)e - 1$, i.e. $g(k) \geq 0$.

(ii) From $x_k = ke - m_k f$ and $x_{k+1} = (k+1)e - m_{k+1} f$ one gets $x_{k+1} - x_k = e + (m_k - m_{k+1})f$. Both x_k, x_{k+1} lie in $[1, f]$, so the left side lies in $[1 - f, f - 1]$; with $0 < e < f$ this forces $m_k \leq m_{k+1} \leq m_k + 1$. Hence $g(k+1) - g(k) = (m_{k+1} - m_k)e - f \leq e - f < 0$.

(iii) Since $1 \leq e \leq f - 1$, the least positive member of the class of e is e itself, so $m_1 = 0$ and $g(1) = 2e - 1 - f$. If $f \geq 2e$ then $g(1) < 0$, and g is decreasing, so $g(k) < 0$ for every $k \geq 1$. In general $g(k) \leq g(1) - (k-1)(f - e)$, and $g(K) \geq 0$ gives the stated bound on K. $\square$

Remark 39 (The wedge-filler hypothesis, and where it fails)**.** Recorded in [**?**]. Full discussion in [**?**].

Lemma 40 (A c-corner carries a side a-edge)**.** [PROVED] *The base corner angle is β, with flanks $\{a, c\}$. If a corner tile lays c on the base it lays a on the side, so that side's a-count P' is positive; by Lemma 94 $P' = fp$, hence $P' \geq f$, and with $pe + R' = f$ and the γ-trap $R' \geq 1$ one gets $1 \leq p \leq (f-1)/e$. So the side condition of Hypothesis 29, namely $p = 0$, fails at any c-corner: under that hypothesis both base ends are a-edges, exactly as Theorem 20 concludes at $e = 1$. What remains for $e \geq 2$ is to exclude $1 \leq p \leq (f-1)/e$, which is the analogue of the entire $e = 1$ programme rather than a single lemma; the boundary alone does not do it, since a and c are both available at the first base position once the walls column is in force.*

Proposition 41 (The chain machinery is member-independent)**.** [PROVED] *Every ingredient of the corner chain holds at all members (e, f), not only at $e = 1$:*

- (i) *the branch relations $3\alpha + 2\beta = \pi$ and $\gamma = 2\alpha + \beta$ define the family, so the unique fills of β, α and $\alpha + \beta$ (`fill_beta`, `fill_alpha`, `fill_alpha_beta`) and the straight and full vertex classifications are unchanged;*
- (ii) *$\cos\gamma = -e/2f < 0$, so $2\gamma > \pi$ and every reflected partner dies at a straight junction, at every member;*
- (iii) *$b \sin\gamma = c \sin\beta$, so an a-up tile spans its strip, at every member;*
- (iv) *for $1 \leq j < f$ the only decomposition of $j\,b$ is j copies of b. Writing $j = y + z + w$, the equation gives $e^2(z+w) + x\,ef = wf^2$, so $f \mid e^2(z+w)$ and hence $f \mid z+w$ by coprimality; since $z + w \leq j < f$ this forces $z = w = 0$, so $y = j$ and $x = 0$. Both reductions are derived rather than assumed (`partition_jb_gen`). Verified for all coprime (e, f) with $e < 12$, $f < 20$.*

The single parameter that changes is the overshoot $c - b = e^2$, which is 1 only at $e = 1$. The $e = 1$ arguments that used $c - b = 1$ as a unit therefore carry over with e^2 in its place, while those that used its being the smallest positive integer do not.

Lemma 42 (The α-vertex gap, general)**.** [PROVED] *A cover piece of a horizontal a-edge has its foot at $(k+1)a$ from the row origin if it is direct, and at $(k+3)a - e^3/f$ if it is mirrored, since $2\,cu_x - 3a = -e^3/f$. The mirrored foot is a lattice point only if $f^2 \mid e^2$, and at row 1, where feet land on the base and every junction there is at an integer abscissa, it is a junction only if $f \mid e^3$; coprimality forces $f = 1$ in both cases (`alpha_vertex_gap_gen`). So the mirrored placement is excluded at row 1 at every member with $f \geq 2$, with e^3 playing the role that 1 plays at $e = 1$.*

Remark 43 (The two corners are not equally rigid)**.** At a c-corner the side junction carries γ and the remaining $\alpha+\beta$ has the unique fill $\{\alpha,\beta\}$, so the partner is forced immediately; this is what makes Theorem 99 short. At an a-corner the side junction carries α and the remaining $2\alpha+2\beta$ admits both $\{2\alpha,2\beta\}$ and $\{\beta,\gamma\}$, so the side branches at once. The base end behaves the other way: there the a-corner leaves $\alpha+\beta$, whose unique fill again forces the partner and confines the next base edge to $\{a,c\}$. The chain therefore runs along the base from either corner, and what it cannot do is pass a base b-edge; this is the mechanism behind Theorem 63 at $e=1$ and it is available verbatim at every member.

Lemma 44 (The corner step, unconditional at every member)**.** [PROVED] *Let a base corner tile lay a on the base. It presents γ at the far end of that a-edge, where the junction is straight, so the remainder is $\alpha+\beta$ with the unique fill $\{\alpha,\beta\}$. The corner tile's b-edge joins a base vertex to a side vertex, so both of its ends lie on the boundary and the tile across it lays a whole b; that tile presents α at the junction, since a straight junction carries at most one γ. Hence the α-tile is adjacent to the corner tile and the β-tile is adjacent to the base, and the second base edge, lying among the β-tile's flanks, is an a or a c.*

Remark 45 (Where the chain stops being free)**.** Recorded in [**?**]. Full discussion in [**?**].

Remark 46 (The thick program)**.** Recorded in [**?**]. Full discussion in [**?**].

Remark 47 (First thick-member certificates)**.** The full $m=1$ search of the $(2,3)$ target ($N=23$, prime, $23\equiv 11 \pmod{12}$) is EXHAUSTED at 19,677 nodes: no tiling exists, and Hypothesis 29 holds at $(2,3)$. The refutation profile differs from the thin members: the semigroup-run prune carries 57% of the kills (against 8% at $(1,5)$), consistent with Lemma 93 strengthening the run arithmetic, and the maximal depth is $5.7f$ against $4.8f$.

Lemma 48 (The crossing kill, and solitude of branches)**.** [PROVED] *A transversal edge crossing a ladder's line within parameter $b+c$ of its start is fatal: the covers cannot cross it, every T-junction forces the next cover edge, and Lemma 79 forbids termination below $b+c$. A branch's own ceilings never trigger this — the k-th strip crossing is endpoint-tangent, sitting exactly $(k-1)c$ east of the corresponding ceiling's end (`crossing_tangency`) — but a* different *same-row branch's ceilings do whenever it sits 1 to $2f-1$ columns west: the offset Df lands the crossing inside a ceiling for every $-2f<D<0$, at parameter b or $2b<b+c$. Hence any strip junction of an A-branch with another A-branch 1 to $2f-1$ columns to its west admits no straddler, and its fills are mates: at most one A-branch per $2f$-column window per row-band carries any straddler at all.*

Lemma 49 (The mirrored piece is charged)**.** [VERIFIED]
A cover piece of a horizontal c-edge is direct, with foot at the row lattice, or mirrored, with foot displaced by $1/f$; the mirrored piece presents β at its left junction, as does its left neighbour, so that junction carries $n_\beta\geq 2$ and its figure is $\{3\alpha,2\beta\}$ (`mirrored_left_junction`). By the census (Lemma 89) each such junction forces an additional $\{\beta,3\gamma\}$ vertex: a single mirrored piece already gives $v_1\geq 2$ (`escape_charge`). The off-lattice escape available at rows $j\geq 3$ is therefore paid for globally, in climber vertices, of which the kernel-checked tilings carry between one and four.

Lemma 50 (Climbers detect deviation)**.** [PROVED] *At an interior lattice point of the brick/mate structure exactly six tiles meet, contributing β,γ,α from the row above and α,γ,β from the row below; the counts are $(2,2,2)$ and the figure is $\{2\alpha,2\beta,2\gamma\}$. Hence a $\{\beta,3\gamma\}$ vertex never occurs where the six incident tiles are the brick/mate six: by Lemma 89 every tiling contains such a vertex, and by Lemma 49 each mirrored cover piece forces a further one, so each escape requires two points at which the tiling deviates from the forced pattern. This* *is a detector, not an exclusion: the kernel-checked tilings do carry climbers at lattice points, at places where their local structure is not brick/mate.*

Remark 51 (What reach buys, and what it does not)**.** Recorded in [**?**]. Full discussion in [**?**].

Remark 52 (The residue at $f \geq 6$)**.** Recorded in [**?**]. Full discussion in [**?**].

Remark 53. The residual class for $f \geq 5$ is precisely: some side's first a-run behind an initial c-block of length at least 2. At $(1, 4)$ the single such side word dies by the line-total kill of Remark 55, whose three gap facts $b - a$, $b - e^2$, $2b - a \notin \langle a, b, c\rangle$ hold for every $f \geq 3$; what does not yet generalize is the case analysis behind them when the initial block exceeds two or the run splits. Closing this class for all f is the last step of Conjecture 60 at $e = 1$.

5. Scope limits and closed routes from the main paper

Remark 54 (Why the colouring method cannot close the rest)**.** The residual gap ($e \geq 2$) is not merely unproved by [**?**, Thm. 14]: it is *unprovable by that method*. Let $\eta = e^{i(\omega+\pi/2)}$, so that $f\eta^2 + e\eta + f = 0$ and every edge direction of every such tiling lies in $\langle \eta, -1\rangle$. Any T-junction-proof invariant of the shape $\sum_{\text{edges}} L \cdot \chi(\theta)$ with $\chi(-w) = -\chi(w)$ — i.e. exactly the Φ/colouring method, and exactly Beeson's — is equivalent to the master identity

$$m\bigl(-c\,z^4 + a\,z^3 + b\,z^2 + a\,z - c\bigr) = z^3(f - ez)P(z) + (fz - e)Q(z), \qquad P, Q \in \mathbb{Z}[z, z^{-1}],$$

whose $z = \pm 1$ specializations return exactly $M = m(f + e)$ and $-m(f - e)$. But that identity is *solvable for every* m, with the explicit witness $P_0 = mf(z^{-1} - z^{-3})$, $Q_0 = m(ez^2 - fz^3)$. Hence no invariant in this class can force $g \mid M$, for any (e, f, m): the Φ-method — the engine of both Beeson's argument and of our own $2\pi/3$ theorem — provably cannot decide the base-β residue. Closing it requires genuinely non-linear input.

The natural non-linear candidate — the full Conway–Lagarias tiling group, whose abelianization is exactly this Φ class — does not help either. For the base-β tile $(2, 3, 4)$ we computed that group over the finite quotients $\mathbb{F}_p[i]$ ($p \equiv 3 \pmod 4$, 15 a quadratic residue), across $p \in \{43, 59, 67, 71, 103, 127\}$: the class-2 (commutator) layer collapses completely onto the tile relators, so the group reduces to its rank-2 abelianization (M_α, M_β), which is admissible for every candidate. In particular the $m = 1$ boundary word for $N = 11$ is trivial in the group — indistinguishable from the $m = 2, 3, 4$ words that *do* tile — while the controls hold (a triangle not congruent to the tile is non-trivial, and the tileable $N = 44, 99, 176$ words are trivial). So the tiling group is not an obstruction here, extending the collapse we observed for the $(8, 7, 13)$ tile (Remark 84) to the base-β tile itself (`code/verify_tiling_group_basebeta.py`). The non-linear input that closes the branch, if any, is not homological.

Finally, the standard *interior* tool fails here too, and for a reason that is an identity rather than an accident. Beeson's Γ_c argument — the engine of his equilateral no-prime theorem — runs by showing Γ_c is empty: if no relation $jc = ua + vb$ is witnessed with j below the tile count, a Γ_c link must emanate from the boundary, which is impossible since links have in- and out-degree 1 and cannot terminate on ∂ABC. For the base-β tile that hypothesis is false uniformly. Reducing $jc = ua + vb$ modulo f twice gives $f \mid v$, then $f \mid u - we$, and the complete solution set

$$v = fw, \qquad u = we + fs, \qquad j = se + wf \qquad (w, s \in \mathbb{Z}_{\geq 0}),$$

generated by

$$e\,c = f\,a \quad (j = e), \qquad\qquad f\,c = e\,a + f\,b \quad (j = f).$$

The first is the identity $ef^2 = f \cdot ef$. So a relation with $j = e$ always exists, and $e < f \ll N = 3f^2 - e^2$; the Γ_c-emptiness contradiction therefore never triggers, for any (e, f). The classification and the witness are machine-checked in Lean (`lean/GammaC.lean`: `gammac_classification`, `gammac_witness`, `gammac_j_lt_N`; axiom-clean). Closing this branch needs control of a *non-empty* Γ_c, which is exactly the step left open in [**?**].

The three failures above share one feature, and that feature settles what any future argument may look like. Fix the tile $(2,3,4)$, which is the base-β tile at $(e,f)=(1,2)$, and consider its targets $(8m,8m,11m)$ with $N=11m^2$. These targets are pairwise similar and the tile is the same in each. The case $m=1$ $(N=11)$ admits no tiling, by exhaustive search; the cases $m=2$ and $m=3$ do admit one, by the kernel-checked certificates of Remark 21. Hence, if $\mathcal{C}$ is any criterion for non-tileability whose hypotheses depend only on the similarity class of the tile and that of the target, taken separately, then $\mathcal{C}$ cannot exclude the base-β target at $m=1$: its hypotheses hold verbatim at $m=2$, where a tiling exists, so $\mathcal{C}$ would contradict it. The statement is formalized and axiom-free (`lean/ScaleRigidity.lean`: `no_similarity_invariant_proof`); only the *existence* at $m=2$ is used, never the search result at $m=1$.

Angle arithmetic, the direction group $\langle\eta,-1\rangle$, the Φ class, the angular layer of the Conway–Lagarias group, and any Dehn-type invariant are all criteria of this kind. So the three obstructions recorded above are not three accidents but one, and the same applies to any further invariant assembled from the same data. What survives is exactly the scale-sensitive information carried by the edge-walk conditions, which do separate the cases: for the tile $(2,3,4)$ every admissible walk on an equal side is free of b-edges when $m=1$, whereas at $m=2$ the walk $(P,Q,R)=(3,2,1)$ is admissible and carries two of them (both machine-checked, ibid.). Any proof of the base-β no-go must take such input.

Remark 55 (A parity argument that fails, and why)**.** It is tempting to argue as follows. Every tile has exactly one b-edge; by Proposition 15 the edge b is never a sum of two or more tile edges; so an interior b-edge is matched by exactly one further b-edge, interior b-edges pair the tiles two at a time, and $N\equiv Q_\partial \pmod 2$ with Q_∂ the number of boundary b-edges. We record that this is *false*, because the middle step does not follow.

Unsplittability forbids an edge from being *exactly partitioned* by two or more tile edges. It does not forbid a *straddle*: a longer edge may contain a b-edge in its interior and overhang it at one or both ends, and then the b-edge has no partner. Indeed the apex-mismatch configuration of [2] is built on precisely such a straddle. The genuine tilings confirm the failure: in the 44-tiling six b-edges belong to a single tile although *no* b-edge lies on the boundary, and the 99-tiling has thirty-three matched pairs, eleven boundary b-edges and twenty-two interior straddles. The correct relation is

$$N = 2\cdot\#\{\text{matched pairs}\} + \#\{\text{boundary}\} + \#\{\text{straddled}\},$$

so the parity conclusion needs the straddle count to be even, which nothing supplies. (It happens to be even, 6 and 22, in these two tilings, which is why the conclusion survives there.)

The same objection defeats the analogous arguments for the a- and c-edges. We state this explicitly because the argument is natural, its conclusion is correct on every example we can test, and it is nevertheless not a proof.

Remark 56 (the tiling group collapses as well)**.** The strongest standard combinatorial obstruction is the Conway–Lagarias *tiling group* $T=\langle\chi_v\rangle/\langle$ collinear additivity; tile boundary words $=1$ in every orientation and reflection $\rangle$; a tileable region R has $\chi(\partial R)=1$ in T. Writing edge vectors as complex numbers in $K=\mathbb{Q}(\sqrt{3},i)$ (rotation by α is multiplication by $\zeta_\alpha=(11+4\sqrt{3}\,i)/13$, by $\frac{\pi}{3}$ by $(1+\sqrt{3}\,i)/2$, reflection by conjugation) and reducing to finite quotients $\mathbb{F}_{p^2}$ with $p\equiv 11 \pmod{12}$ — so $\sqrt{3}\in\mathbb{F}_p$ but $i\notin\mathbb{F}_p$ and the plane stays genuinely two-dimensional — the tile relators, *together with* the commutators [class-1(tile), generator] that normal closure forces (a tile word's class-1 image is itself a relator, hence not central), fill the entire commutator layer. So T collapses to its rank-two abelianization (M_α,M_β): the class-2 quotient is trivial while the class-1 layer keeps rank 2. As $N = 105$ is admissible, $\chi(\partial_{105}) = 1$, and the tiling group does not obstruct it. We verified this exactly for $p \in \{23, 47, 59, 71, 83, 107, 131\}$ (`verify_tiling_group.py`), and — extending to the free nilpotent quotient through class 4 and to a range of finite non-abelian groups (through S_5, $\mathrm{GL}_2(\mathbb{F}_p)$, $\mathrm{PSL}(2,7)$ and the relevant Frobenius groups, which carry genuine non-abelian data: sensitivity controls fire on 92–96% of tile-consistent representations) — found *no* tile-consistent quotient sending ∂_{105} off the identity, with the reptile controls $N = 4, 9$ trivial throughout and a Φ-nonadmissible loop not. The no-go of Remark 83 therefore extends past the linear invariants to the full tiling group: on this tile the tiling-group obstruction is exactly Φ, so deciding realizability of $N = 105$ lies outside every standard tiling invariant.

Remark 57 (The scalene $2\pi/3$ shape F_4 first becomes possible at exactly 88)**.** The sweeps of Theorems 67–77 test the isosceles and F_1 targets among the $2\pi/3$ shapes and omit the scalene families F_2, F_3, F_4, on the ground that their minimal counts exceed the range under consideration. That ground is exact, and it expires inside the present band. Writing $N = k^2 N_0$ with N_0 the closed-form ratio of Section 7, the minima over primitive 120°-triples are

$$\min N_0(F_2) = 143 \text{ at } (a,b) = (3,5), \qquad \min N_0(F_3) = 264 \text{ at } (5,3), \qquad \min N_0(F_4) = 88 \text{ at } (3,5),$$

so in $N \le 90$ these three families contribute exactly one instance, and it occurs at the endpoint $N = 88$: the tile $(3,5,7)$, with its $2\pi/3$ angle opposite 7, on the scalene target $(21, 55, 56)$, at $k = 1$. The area ratio is exact, $660\sqrt{3} = 88 \cdot \frac{15\sqrt{3}}{2}$. It is the first instance from any of F_2, F_3, F_4 to enter the frontier sweep — scalene targets themselves are not new, the sweep of Section 7 having already searched the same tile $(3,5,7)$ on $(15, 21, 24)$ at $N = 24$ and $(7, 8, 13)$ on $(56, 104, 120)$ at $N = 120$ — and it is the reason 88 is not settled by the four exhaustions above. The instance is under search. The bound also confirms that no scalene $2\pi/3$ shape can affect any $N < 88$, so Theorem 77 and everything below it are untouched.

Remark 58 (The third isosceles base-α instance)**.** The isosceles base-α shape of [**?**, Thm. 19] must be enumerated *without* that theorem's divisibility condition $g \mid M$, whose proof is unsound (Remark 11): imposing it reports the branch empty at $N = 21$ and $N = 70$, whose instances are real and needed the engine. Solving the branch's defining cubic $N/M^2 = (1+s)(2-s^2)/\big((1-s)(2+s)^2\big)$ over the rationals in $0 < s < 1$, the solutions with $N \le 90$ are exactly three:

$$(N, M, s) = (21, 5, \tfrac{1}{2}), \qquad (70, 8, \tfrac{2}{3}), \qquad (84, 10, \tfrac{1}{2}),$$

the first two being the instances of Theorem 67 and Remark 73, exhausted at 13 659 and 134 631 158 nodes. The third is new and is the only one above 80. Since $s = e/f$ it forces $(e, f) = (1, 2)$ and hence the tile $(2, 3, 4)$ — the same tile as the 44-tiling — on the isosceles $(24, 24, 42)$ with base angles α, apex at $(21, 3\sqrt{15})$. So 84 carries three instances, not the two the $(a+b)$-shape sweep produces: $(10, 21, 25)$ on $(210, 210, 84)$, already exhausted at 78 nodes, and $(110, 21, 121)$ on $(462, 462, 420)$, exhausted at 4 347 701 nodes, leaving only $(2, 3, 4)$ on $(24, 24, 42)$ under search. Sparse though this branch is — three instances in ninety values — it is not exhausted by them, which is why the band cannot be closed by N-forms alone.

Remark 59 (no invariant decides realizability)**.** The directional invariants are *conserved* by the tiling process: each tile carries value $\pm(c + a - b)$ for f_α (resp. $\pm(c + b - a)$ for f_β), so placing one tile changes Φ_f of the unfilled region by $\pm V$ and the number of remaining tiles by one — in step, so that Φ_f / V stays integral and keeps the parity of the tile count throughout. Hence the only conditions such an invariant imposes are the admissibility conditions of Theorem 56, and *no* additive boundary invariant can distinguish an admissible target from a partially tiled one; none can obstruct an admissible instance, and none prunes the search below its combinatorial size. For the sharpest open instance $N = 105$ we confirmed this computationally: searching all directional weights of period ≤ 4 in the α-winding (reflections included), the only invariants are

f_α ($M_\alpha = 5$) and f_β ($M_\beta = 21$), both admissible; and the vertex-type balance forced by the angle relations $a\alpha + b\beta + c\gamma \in \{\pi, 2\pi\}$ admits 42,282 nonnegative integer solutions. Deciding realizability of an admissible instance therefore lies beyond every linear or coloring invariant; it is a genuinely combinatorial question, which is why the search of Theorem 82 appears to be essential.

Remark 60 (The spectrum is member-dependent)**.** Theorem 5 does not extend along the base-β family. For $(e, f) = (1, 3)$ — tile $(3, 8, 9)$, $N_1 = 26$ — the scale-2 target admits no tiling: $N = 104$ is excluded by exhaustive search (43 541 916 nodes). The two ingredients of the induction therefore behave differently along the family: the unit parallelogram is constructible for every member with $e = 1$ (companion note, Theorem on the unit parallelogram; verified for $f = 2, \dots, 12$), while the seeds Δ_2, Δ_3 that the collar needs are not. Every attempt to build the seed at $f = 3$ fails for the same reason, namely that there is nothing to build: the cevian piece, the symmetric middle piece, and the maximal interiors of both natural lattices all return NO TILING, and a seeding by unit parallelograms is impossible by counting. Consistently with this, $(1, 2)$ is the unique member of its family with $\beta < \pi/3$. What the spectrum of $(1, 3)$ is remains open: since Δ_4 would require $\Delta_1 + \Delta_3$ and Δ_5 would require $\Delta_2 + \Delta_3$, both excluded, the set reachable from Δ_3 alone is exactly the multiples of 3.

Remark 61 (Why the forms, and what the reformulation buys)**.** Every tile count in the classification is a value of a binary quadratic form: $e^2 + f^2$ (discriminant -4) on the commensurable branch, the Eisenstein norm a^2+ab+b^2 (discriminant -3) for a $2\pi/3$ tile, $3f^2 - e^2$ (discriminant 12) and $2f^2 - e^2$ (discriminant 8) on the $3\alpha + 2\beta$ branch. The classical exclusion of primes $\equiv 3 \pmod 4$ on the commensurable branch is exactly Fermat's two-squares theorem for the first form; Theorem 4 is its analogue for the third. Two consequences. First, the exception in Theorem 1 is not a Diophantine side condition but a single arithmetic progression, and by Dirichlet it carries half of the primes $\equiv 3 \pmod 4$: the theorem is unconditional on the other half. Second, $19 \equiv 7 \pmod{12}$, so the value in Erdős's question lies in the unconditional class. The forward implication is machine-checked (`basebeta_prime_mod_twelve`, `BaseBetaMod12.lean`); it is the direction the corollary uses. The converse was checked for all 109 primes $\equiv 11 \pmod{12}$ below 3000 with no exception.

Remark 62 (What 83 rests on)**.** Among the values of the band, 83 is of a different kind from the rest, and the distinction is sharper than mere corroboration. Every prime $\equiv 11 \pmod{12}$ below it — $11, 23, 47, 59, 71$ — was settled by an independent certified exhaustive search of its base-β instance, so those exclusions stand on the engine alone and need no branch theorem. For 83 the single surviving instance is the tile $(1722, 83, 1764)$ on $(3486, 3486, 3403)$, the base-β member $(e, f) = (5, 6)$, and that search has not terminated. Its exclusion therefore rests entirely on Theorem 2, which is conditional on the companion's complete-corner-wall hypothesis (Remark 3). *So* 83 *is not excluded unconditionally by this paper*, and it is the first value for which that is true: everything below 81 is unconditional, either by an N-form, by Theorem 1 for primes $\equiv 7 \pmod{12}$, or by an exhaustion. We state this explicitly because the boundary between the two kinds of result is easy to lose in a table, and because the same hypothesis governs every $N \equiv 11 \pmod{12}$ above 83 as well.

Remark 63 (What this does and does not remove)**.** Proposition 37 is a statement about the $2\pi/3$ branch only. It says nothing about the base-β branch of $3\alpha + 2\beta = \pi$, which is a different branch with a different tile and is where the one genuine gap of Theorem 1 lives (Remark 13). What it does buy is a narrowing of the rationality dependence: Corollary 38 and Proposition 39 discharge three of the eleven shapes — tile-similar, F_1, F_1' — with no appeal to [**?**] or [**?**, Thm. 12.5], whereas the remaining eight still use an integer-sided tile. The obstruction to doing the same for F_2, F_3, F_4 is visible in the table: there $\kappa \neq \pm 1$, so the product pins only the single ratio a/b, and the law of cosines then leaves c/b a quadratic irrationality that no further invariant condition controls. Establishing rationality on a general target remains exactly the content of the cited theorem.

Remark 64 (Scope)**.** The hypothesis is explicit and strictly weaker than the imported theorem: excluding ($\star$) is a corner-and-walk exclusion of the kind carried out for the base-β target, where the γ-trap shows every side carries a c-edge. It is not established here for these six shapes. The remaining two shapes are unaffected: on F_2 the same elimination gives a quadratic in a/b with discriminant $9\kappa^2 - 30\kappa + 9$, and on the equilateral target $\kappa = 3$ is constant, so both continue to use [**?**]. The arithmetic is machine-checked in `lean/RationalityFree.lean`. That the condition has content is visible from the fact that $a/b \in \mathbb{Q}$ alone does not give $c/b \in \mathbb{Q}$: the latter asks that $(a/b)^2 + (a/b) + 1$ be a rational square, which holds for $a/b = 3/5$ and $7/8$ but not for $a/b = 1$ or 2; the tiles for which it holds are the integer triangles with a $2\pi/3$ angle.

Remark 65 (Scope)**.** The proposition converts the rationality input from a hypothesis about the tile into a hypothesis about the invariant, and the latter is what the method produces natively: wherever the boundary functional of Section 4 is defined and integral, rationality is automatic. It does not by itself discharge the citation on the *general* target, because establishing integrality of all three quantities there is exactly the content of the cited theorem. What it does discharge is the appearance of a rationality assumption *inside* the invariant argument, where the three integralities are already in hand; and it shows the two facts are equivalent rather than independent, which is not how the literature presents them.

Remark 66 (Scope)**.** Two points. First, the exclusion of 11 together with the realizability of $11 \cdot 2^2$ and $11 \cdot 3^2$ shows that a base-β family can be empty at scale 1 and non-empty above it; the obstruction at $m = 1$ is genuinely a small-scale phenomenon and not a property of the family. Second, before this the smallest tiling known in any incommensurable branch had 1215 tiles; the corollary supplies infinitely many, of which 44 is the smallest. Whether $S(e, f) \neq \emptyset$ for every (e, f) — equivalently, whether every base-β family tiles at *some* scale — is open; the first untested instances are the scale-2 targets of the families $(2, 3)$, $(3, 4)$ and $(1, 4)$, with $N = 92$, 156 and 188.

Remark 67 (The golden ratio)**.** Two consequences. First, Proposition 15 is exactly the hypothesis the corner parallelogram needs, so Proposition 28 holds with *no* proviso. Second, $e = 1$ — which both genuine tilings realize — is the only case in which an edge splits at all, while the open $e \geq 2$ regime is rigid. The comparison of a with b must be made with care: $b > a \iff f^2 - ef - e^2 > 0 \iff f/e > \varphi$, the golden ratio. The thick regime straddles that threshold: $N = 66$ $(3, 5)$ and $N = 131$ $(4, 7)$ are super-golden, while $N = 23$ $(2, 3)$, $N = 59$ $(4, 5)$, $N = 83$ $(5, 6)$ and $N = 167$ $(5, 8)$ have $b < a$. Proposition 15 above assumes no inequality between a and b.

Remark 68 (What the alignment pins, uniformly)**.** For every open thick member the mismatch ray is now completely determined out to length $f \cdot b$: its far wall is b^f, terminating transverse to the opposite side, and $V = f^2$ falls strictly inside the $\lceil f^2/b \rceil$-th b-edge. Explicitly (all exact): $N = 23$ far side $b^3 = 5^3$, $V = 9 \in (5, 10)$; $N = 59$ $b^5 = 9^5$, $V = 25 \in (18, 27)$; $N = 66$ $b^5 = 16^5$, $V = 25 \in (16, 32)$; $N = 83$ $b^6 = 11^6$, $V = 36 \in (33, 44)$. This pins an extended *interior* structure of a hypothetical thick tiling, uniformly in (e, f); it serves as a root prune for the apex-down search; the forced b^f wall must land on the base at coordinate $e \cdot f^2$, whose b-count is bounded (`base_b_bound`). It does not, on its own, close any member.

Remark 69 (Scope)**.** The irrationality hypothesis cannot be dropped: the commensurable tile $(\frac{\pi}{6}, \frac{\pi}{6}, \frac{2\pi}{3})$ tiles an equilateral (hence isosceles) triangle with $N = 3$, by the fan from the centre;

that tile lives in the commensurable branch of the classification, where $N = 3n^2$ is allowed. The equilateral target for an *incommensurable* $2\pi/3$ tile is excluded for primes by Beeson [**?**], and is covered in the assembly of Section 6; it is not part of this theorem.

Remark 70 (Scope)**.** The hypothesis that the corner angles are $\beta, \beta, 3\alpha$ is essential, and this is why [**?**, Lem. 6] is false as quoted (Remark 9): on the tile-similar target a corner *is* a γ, the chain terminates harmlessly, and the counterexample there has two sides with no c-edge. On a base-β target no corner is a γ, and the argument runs.

Remark 71 (The field of computation)**.** For the tile (ef, f^2-e^2, f^2) one has $\sin\alpha = e\sqrt{D}/(2f^2)$ with $D = 4f^2 - e^2 = (2f-e)(2f+e)$, so a base-$\beta$ tiling's coordinates lie in $\mathbb{Q}(\sqrt{D})$. This is the field the exact search computes in; for $(e, f) = (1, 2)$ it is $\mathbb{Q}(\sqrt{15})$, the ring of all three tiling certificates.

Remark 72 (Validation on the 44-tiling)**.** On the genuine 44-tiling ($f^2 = 4 > 2e^2 = 2$, so $t = 0$) the piercer spans exactly $[3, 6]$, starting at the T-junction, with corner α there to numerical precision, as forced; the arithmetic core is machine-checked (`pre_pierce_dichotomy`). The rung also has a computational consequence: the apex neighbourhood is rigid (three forced α-tiles, a forced mismatch ray, a forced T-junction, a forced piercer start), while the base admits several walks. Since the corner-anchored search anchors at the lowest vertex, orienting the target apex-down (a reflection, under which verdicts are preserved) places the rigid chain at the root of the search tree: on $N = 23$ the apex-down search exhausts in 3,954 nodes against 19,677, whereas on the thin member $N = 47$, where the base is the rigid side, the apex-down orientation is slower. The thick searches $N = 59, 66$ are run apex-down.

6. The crossing question is the single obstruction

The preceding sections record failures one route at a time. This section states the consolidated fact they add up to, which is sharper than any of them separately: every route to the base-β primes known to this programme reduces to one question, and every invariant class available to the field is provably unable to answer it.

Definition 73 (The crossing question)**.** Let (e, f) be coprime with $1 \le e < f$, let $\mathcal{T}$ be a tiling of the base-β target at $m = 1$, and let S be a maximal run of forced tiles whose far sides lie along a segment L of a line (a corner chain along a wall, or a column footprint along a level line), with V the endpoint of L beyond the run's last forced tile. The question: *must the next tile along L lay a whole edge on L without overrunning V, or can it cross L at V?*

Proposition 74 (Three reductions)**.** [PROVED] (i) *At $e = 1$ the entire subfamily — every prime $3f^2 - 1$ — reduces to the instance of the crossing question at the column footprint: the forced column exists in every hypothetical tiling (both corners break; a base-break consumes the single c), its descent is decided by the length arithmetic except at the crossing, and a landing closes the family (*`A2BranchRow3.row_three_dies_of_span`*,* `span_all_a`*).* (ii) *At $e \ge 2$ the walk narrowing of Corollary 100 rests on Theorem 99, whose induction fails at the interior points V_k (Remark 4); Remark 5 shows the missing step* is *the crossing question at V_k, verbatim.* (iii) *Conjecture 60 is the crossing question by construction. No fourth wall exists: the wedge-filler condition, once conjectured to block the regime $e^4 - 4e^2f^2 + 2f^4 \le 0$, blocks nothing (the fill is forced at every member by the integrality of the vertex system;* `PrimeRegimes`*).*

Proposition 75 (Nine tool classes cannot answer it)**.** [PROVED]
Each of the following is closed with a recorded failure point: linear direction invariants and their metric bounds (`LambdaFactor`*); Euler and angle-sum counting, which coincide in a single relation (§4); edge-pairing parity; the non-abelian Conway–Lagarias tiling group, which collapses*

to its rank-two abelianisation (Remark 56); every criterion built from similarity-invariant data of tile and target separately (`ScaleRigidity`, using only the existence of the $m = 2$ tiling); the γ-count on a stretch, whose two-endpoint hypothesis is unsatisfiable under the monotone orientation word; the b-residue calculus, which constrains counts exactly and locations not at all (58 of 60 b-edges in the certificates have an endpoint that is not a junction of the far side); De Bruijn–Hamel functionals, which require metric incommensurability while every side of the family is an integer; and equilibrium stresses, whose balance equations have slack at least two at every vertex type of the configuration, so that no local forcing exists.

The common feature is exact: each tool computes a quantity invariant under relocating edges within the tiling, and the crossing question asks *where* an edge lies. The searches, which do decide members, decide them precisely because a search tracks locations. Any future attack should be measured against this requirement before effort is spent.

7. Verification, claim by claim

The finite and numerical claims are checked by the scripts in the repository (exact arithmetic). `verify_shapes.py` reproduces the eleven-shape enumeration of Section 2 and the values N_0 of Section 3. `verify_invariant.py` confirms Lemma 35 ($C_f(t) = \pm(c+a-b)$ over all orientations of T), Lemma 34 (both sides agree on explicit subdivision tilings, and (the antisymmetry identity of [1]) cancels at a non-edge-to-edge incidence), and Lemma 41 (no counterexample among primitive 120°-triples up to $c \approx 9 \times 10^6$). `verify_spectrum.py` checks Section 7: the two-positive-squares decomposition for all primes $p \not\equiv 3 \pmod 4$ below 10^5; the scale structure $b \mid k^2 \iff de \mid k$; the admissible spectrum of Theorem 53 (no prime is admissible; 33 and 46 are); and that every instance of $e \mid (a + b - c)$ with $a, b < 3000$ arises from the j-classification of Proposition 54, with none for $d = 1$. As a positive control, both necessary conditions of Theorem 42 hold on the genuine 2673-tile isosceles tiling of Herdt [?]: for its tile $(5, 3, 7)$ and scale $k = 27$, $N = k^2(a + 2b)/b = 2673$ and $M = k(2b + a - 2c)/(c + a - b) = -9$ are integers, as they must be on any true tiling.

The *geometric* layer is also machine-checked, end to end and with no prose joints. That the angles of the tiles meeting an interior point sum to 2π — the fact a dissection argument uses constantly and which has no counterpart in Mathlib — is obtained without any theory of angular measure, by a chain that is purely measure-theoretic: `TangentCone.poly_inter_ball_eq_coneAt` (a polytope agrees with its tangent cone at a point inside a small ball), `SectorArea.volume_sector` (the unit sector of angle θ has area $\theta/2$, via polar coordinates), `Wedge.sector_eq_halfplanes` (that polar sector is exactly the intersection of two half-planes with the unit disc), `E2Join.volume_halfplane_wed` and `E2Join.volume_wedge` (the same area statement transported to `EuclideanSpace` $\mathbb{R}$ `(Fin 2)`), and finally `AngleSumAssembled.angle_sum_interior`. Every step is axiom-clean.

The entire *arithmetic and combinatorial* layer of the proof is formalized and machine-checked in Lean 4 with Mathlib (`lean/Erdos634/PrimeArithmetic.lean`, twenty-three theorems); every theorem depends only on the standard axioms `propext`, `Classical.choice`, `Quot.sound`, with no `sorry`:

> `shape_enumeration`: the eleven-triple completeness of Section 2 — realizable corner types with $\sum m_i = 0$, $\sum k_i = 3$ are exactly the eleven listed — as pure integer arithmetic;
> `k_not_dvd_sum_sub`, `M_not_int`: Lemma 41 ($(c - a - b)/\sqrt{b} \notin \mathbb{Z}$);
> the full arithmetic of Theorem 42 in one master statement: no prime N satisfies the 120°-relation, the area equation $Nb = k^2(a + 2b)$, and the integrality $(c + a - b) \mid k(2b + a - 2c)$ simultaneously (`iso_reduction_identity`, `prime_count_forces_scale`, `no_prime_isosceles_count`);

`add_not_prime`, `not_prime_of_two_le`, `F1_count_not_prime`–`F4_count_not_prime`: the scalene counts of Proposition 33 are composite ($a+b$ is never prime for a primitive $120°$-triple, and the F_2, F_3, F_4 counts factor);
`prime_three_mod_four_excluded`: a prime $p \equiv 3 \pmod 4$ with $p > 3$ is neither a square, a sum of two squares, nor $2n^2$, $3n^2$ or $6n^2$ (the commensurable branch of Section 2), via Fermat's two-squares theorem;
`prime_sum_two_pos_squares`: a prime $p \not\equiv 3 \pmod 4$ is a sum of two *positive* squares (the achievability half of Theorem 51);
`iso_admissible`: Theorem 53 — given $b = de^2$ with d squarefree, the area equation and Φ-divisibility force $k = dew$, $N = dw^2(a+2b)$ and $e \mid w(c-a-b)$;
`F1_product`, `F1_Ma`, `F1_Mb`, `F1_ratio`, `F1_pin`, `disc_not_square`, `tile_similar_not_prime`: the arithmetic of Proposition 37, Corollary 38 and Proposition 39 (`lean/InvariantProduct.lean`);
the finite arithmetic of the frontier sweeps, in eight theorems: the $\pi/3$-equilateral square criterion fails at 14 and 15; the first tiling equation has no admissible solution there; the two equilateral factor-pair uniqueness statements; the boundary congruences that kill the isosceles candidates at 14 and 22; the F_1 invariant kill at 21; and the parity kill of 46.

Mathlib still has no theory of planar dissections — no `IsTiling`, no planar subdivision, no Jordan curve theorem, no Euler formula, no polytopes — so the geometric layer is built from the convexity and measure-theoretic primitives that do exist. Within that constraint it is now substantially machine-checked, and we state exactly how far, since the boundary has moved.

Formalized (`lean/Dissection.lean`). A dissection is defined as a target, N tiles, the covering equation, and pairwise disjointness of tile *interiors* — combinatorially, not measure-theoretically, so that the measure-theoretic consequence is derived rather than assumed. From it: `Dissection.aedisjoint` (disjoint interiors give a.e. disjointness, via `Convex.addHaar_frontier`; this is what lets a *non-edge-to-edge* dissection still be additive for area), the area identity `Dissection.volume_target`, and `vertex_multiplicities`, which converts a real angle equation at a vertex into the integer multiplicity equations the arithmetic consumes.

Also formalized: that each side of the target is a finite union of whole tile edges (`hasEdgeChains_edge`), by way of a support-face argument (`SupportFace.mem_convexHull_max`: a point of a simplex attaining the maximum of a linear functional is a convex combination of the vertices attaining it) together with a density step (`SegmentDense.subset_closure_diff_finite`).

Lemma 34 was the one place where geometry remained, and it is now discharged. `InvariantCore.cancellatio` derives the lemma from a single hypothesis: that the total directed length of interior tile edges is invariant under reversal of direction. That hypothesis is no longer assumed. `Dissection.g4_final` proves it for every dissection, unconditionally.

The route avoids the notion of a maximal interior segment, for which Mathlib offers no support. Instead of covering each maximal segment once from each side, one takes $S(d)$ to be the union of the interior tile edges carrying unit direction d, and shows that $S(d)$ and $S(-d)$ agree away from the finite set of tile vertices. The measure-theoretic ingredients are: a tile coincides with a half-plane near a point in the relative interior of one of its edges (`Tri.inter_ball_eq_halfplane`); a half-plane through the centre halves a ball (`volume_halfspace_inter_ball`, which Mathlib does not have); a ball inside the target is partitioned by the tiles (`Dissection.volume_ball_eq_sum` and the resulting count forces exactly two tiles at each such point (`Dissection.two_tiles_at_edge_point`), the classification of a tile's local contribution being exhaustive away from the tiles' vertices (`Tri.classify`), a finite set.

The orientation step carries the genuine geometric content, and runs as follows. The two tiles have disjoint interiors, so the open half-planes they occupy near the shared point are disjoint

(`Dissection.cross_disjoint_of_onEdge`, which uses the structural disjointness of interiors rather than any measure); two nonzero linear functionals whose positive sets are disjoint are negative multiples of one another (`proportional_of_disjoint_pos`); hence the two edges span the *same* line with *opposite* senses (`Dissection.leftDir_antiparallel`); and normalising to unit directions yields $S(d)\backslash F = S(-d)\backslash F$ with F the finite vertex set (`Dissection.dirSet_horient`). Every statement named in this paragraph depends on `propext`, `Classical.choice` and `Quot.sound`, and on nothing else.

What remains on written proofs, checked numerically but not formalized, is the direction grid of Section 2 and Lemma 35. Search verdicts (`EXHAUSTED`) are declared computer assistance and are not formalized: doing so would mean verifying the search engine. Finally, a formalized theorem certifies its own Lean statement; that each such statement faithfully renders the paper result it is cited against is prose, and should be read as such.

Acknowledgements

This work was carried out with substantial assistance from an artificial-intelligence system (Anthropic's Claude), which proposed the signed-direction invariant, performed the symbolic and numerical verification, found the elementary proof of Lemma 41, and produced the Lean formalization, under the author's direction and review. The argument depends on the cited classification of Laczkovich and Beeson for the branches other than the $2\pi/3$ tile, and on the rationality theorem of Beeson and Zhang; the shape list coincides with Zhang's six sporadic types [**?**]. The present contribution is the invariant of Section 4, the consequent exclusion of *every* prime for the isosceles $2\pi/3$ branch (where Beeson had ruled out only $N < 36$ and left primality open), and the self-contained reduction of the scalene types. Its arithmetic layer is machine-checked, and so now is the geometric layer, the interior angle sum included (Section 8). What is *not* settled is the base-β branch: Theorem 2 is conditional on the complete-corner-wall hypothesis of the companion note, and Remark 3 states precisely what survives without it. The whole is offered for independent refereeing.

References

[1] V. Bonfioli, *Erdős Problem 634: the prime case*, 2026.
[2] V. Bonfioli, *Erdős Problem 634: the base-β branch*, 2026.